(SSUET/QR/111)

**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY**
**COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT**
**BS COMPUTER SCIENCE**

# COURSE INFORMATION SHEET

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| **Session:** | Fall 2023 |
| **Course Title:** | Differential Equations |
| **Course Code:** | MS-104 |
| **Credit Hours:** | 3 |
| **Semester:** | $3^{rd}$ |
| **Pre-Requisites:** | MS-103 Calculus & Analytical Geometry |
| **Instructor Name:** | Maria Rashid |
| **Email and Contact Information:** | mariar@ssuet.edu.pk (03312203910) |
| **WhatsApp Group** | MS-104 DE |
| **Office Hours:** | 09:00 AM to 5:00 (Monday to Friday) |
| **Mode of Teaching:** | Synchronous/Asynchronous/**Hybrid (Blended)** |

**COURSE OBJECTIVE:**

The objective of this course is to introduce students the fundamental concepts of Differential Equations, providing the theoretical knowledge and applications in Electronic Engineering.

**COURSE OUTLINE:**

Introduction of Differential equations, Ordinary and Partial Differential equations order, degree and linearity, formation and solution of Differential equations, General and Particular Solutions, Initial and Boundary Value Problems, Separable equations, Equations reducible to Separable Form, Homogeneous Differential Equations, Reducible to Homogeneous equations, Exact Differential Equations, Integrating Factors of Non-Exact Differential Equations, Linear Equations, Bernoulli equations, Orthogonal Trajectories in Cartesian and polar coordinates.
Higher Order Homogeneous & Non-Homogeneous Linear D.E. with Constant Coefficients, Auxiliary Equations, Complimentary functions, Variation of Parameter, Underdetermine Coefficient.
Basic concept and formation of PDEs, Linear Homogeneous PDEs, Lagrange solution of $1^{st}$ order Linear Homogeneous PDEs, D' Alembert's solution of wave equation, Dimensional Wave equations.
Periodic Functions, Expansion of Periodic functions in Fourier series, Fourier Coefficients, Odd and Even Functions and their Fourier series (Fourier Sine and Cosine Series), Fourier series of functions with arbitrary periods, Half-Range Expansions of functions with arbitrary periods.

**COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes (PLOs):**

| CLO No. | Course Learning Outcomes (CLOs) | PLOs | Bloom's Taxonomy |
|---|---|---|---|
| 1 | **Understand** the concept of ordinary Differential equations (first order) **application** in engineering problems. | PLO_1 (Knowledge for solving computing problem) | C3 (Applying) |
| 2 | **Discuss** various techniques of Fourier series and higher order ordinary differential equation. | PLO_1 (Knowledge for solving computing problem) | C2 (Understanding) |
| 3 | **Understand** and **apply** the concept of Partial Differential Equations in engineering problems. | PLO_2 (Problem Analysis) | C3 (Applying) |

**RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:**

| Assessment Tools | CLO-1 | CLO-2 | CLO-3 |
|---|---|---|---|
| **Quizzes** | 3 (6.25)% | 3 (13.63)% | 4 (13.33)% |
| **Assignments** | 3 (6.25)% | 3 (13.63)% | 4 (13.33)% |
| **Midterm Exam** | 20 (45.83)% | 10 (26.67)% | -------- |
| **Final Exam** | 08 (41.67)% | 22 (26.67)% | 20 (73.33)% |
| **Total** | 34 | 38 | 28 |

**GRADING POLICY:**

| Assessment Tools | Percentage |
|---|---|
| Quizzes | 10% |
| Assignments | 10% |
| Midterm Exam | 30% |
| Final Exam | 50% |
| TOTAL | 100% |

**RECOMMENDED BOOKS:**

- Dennis G. Zill, **"A First Course in Differential Equations with Modeling Applications Solutions Manual 11 ed"** 2017 ISBN: 9781305965720

**REFERENCE BOOKS:**

- Richard Bronson & Gabriel Costa, **"Schaum's outline of differential equations,4th Edition"** 2014 ISBN:9780071824859
- Erwin Kresyzig, **"Advanced Engineering Mathematics, 10th Edition"** 2011 ISBN:9780470913611

# LESSON PLAN

**Course Title:** Differential Equations
**Course Code:** MS-104

| Week No. | Week Dates | Topics | Required Reading | Key Date |
|---|---|---|---|---|
| **1** | 09-10-2023 to 13-10-2023 | Introduction of Differential equations: Ordinary and Partial Differential equations.<br>Order, degree, linearity, formation and solution of DEs, General and Particular Solutions, Initial and Boundary Value Problems. | Chap 1 (Page 01-35) | |
| **2** | 16-10-2023 to 20-10-2023 | Homogenous and reducible to homogeneous equations, Separable equations, Separable variable method of Differential equations. | Chap 2 (page 46-54) (page 72-73) | |
| **3** | 23-10-2023 to 27-10-2023 | Equations reducible to Separable Form and their solutions | Chap 2 (page 46-54) | Assignment # 01 |
| **4** | 30-10-2023 to 03-11-2023 | Exact Differential equations, Exact DEs, Integrating Factors of Non-Exact DEs. | Chap 2 (page 64-71) | |
| **5** | 06-11-2023 to 10-11-2023 | Linear Differential Equations.<br>Equations reducible to linear DE<br>Bernoulli equation | Chap 2 (page 55-63) (page 72-76) | Quiz # 01 |
| **6** | 13-11-2023 to 17-11-2023 | Orthogonal Trajectories in Cartesian and polar coordinates. | Chap 1 (Kreyszig) (page 36-38) | |
| **7** | 20-11-2023 to 24-11-2023 | Higher order Homogeneous and Non-Homogeneous Linear Des: Homogeneous Linear DEs:<br>Reduction of order technique | Chap 4 (pg 119-134) | Assignment # 02 |
| **8** | 27-11-2023 to 01-12-2023 | General Solutions of Non-Homogeneous Linear DEs with constant coefficients by the method of Variation of Parameters. | Chap 4 (pg 135-140) (pg 159-165) | |

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| **9** | **Midterm Examination (04-12-2023 to 09-12-2023)** | | | |
| **10** | 11-12-2023 to 15-12-2023 | Euler- Cauchy equations and their solutions by Variation of parameter method | Chap 4 (pg 166-172) | |
| **11** | 18-12-2023 to 22-12-2023 | Periodic functions & expansion of periodic functions in Fourier series, Fourier Coefficients. | Chap 11 (Kreyszig) (pg 474-491) | |
| **12** | 25-12-2023 to 29-12-2023 | Odd and Even Functions and their Fourier Series (Fourier Sine and Cosine Series). | | |
| **13** | 01-01-2024 to 05-01-2024 | Fourier Series of functions with arbitrary periods.<br>Half-Range Expansions of functions with arbitrary periods. | | Quiz # 02<br>Assignment # 03 |
| **14** | 08-01-2024 to 12-01-2024 | Basic concept and formation of PDEs.<br>Linear Homogeneous PDEs.<br>Lagrange solution of 1st order Linear Homogeneous PDEs. | Chap 8 (Potter) (pg 453-459) | Quiz # 03 |
| **15** | 15-01-2024 to 19-01-2024 | Reduction to ODEs and separable variable methods. | Chap 8 (Potter) (pg 476-481) | |
| **16** | 22-01-2024 to 26-01-2024 | D'Alembert's solution of the wave equation<br>Revision | Chap 8 (Potter) (pg 470-473) | |
| **Final Examination (31-01-2024 to 10-02-2024)** | | | | |